WideEP Fault Tolerance in TensorRT-LLM

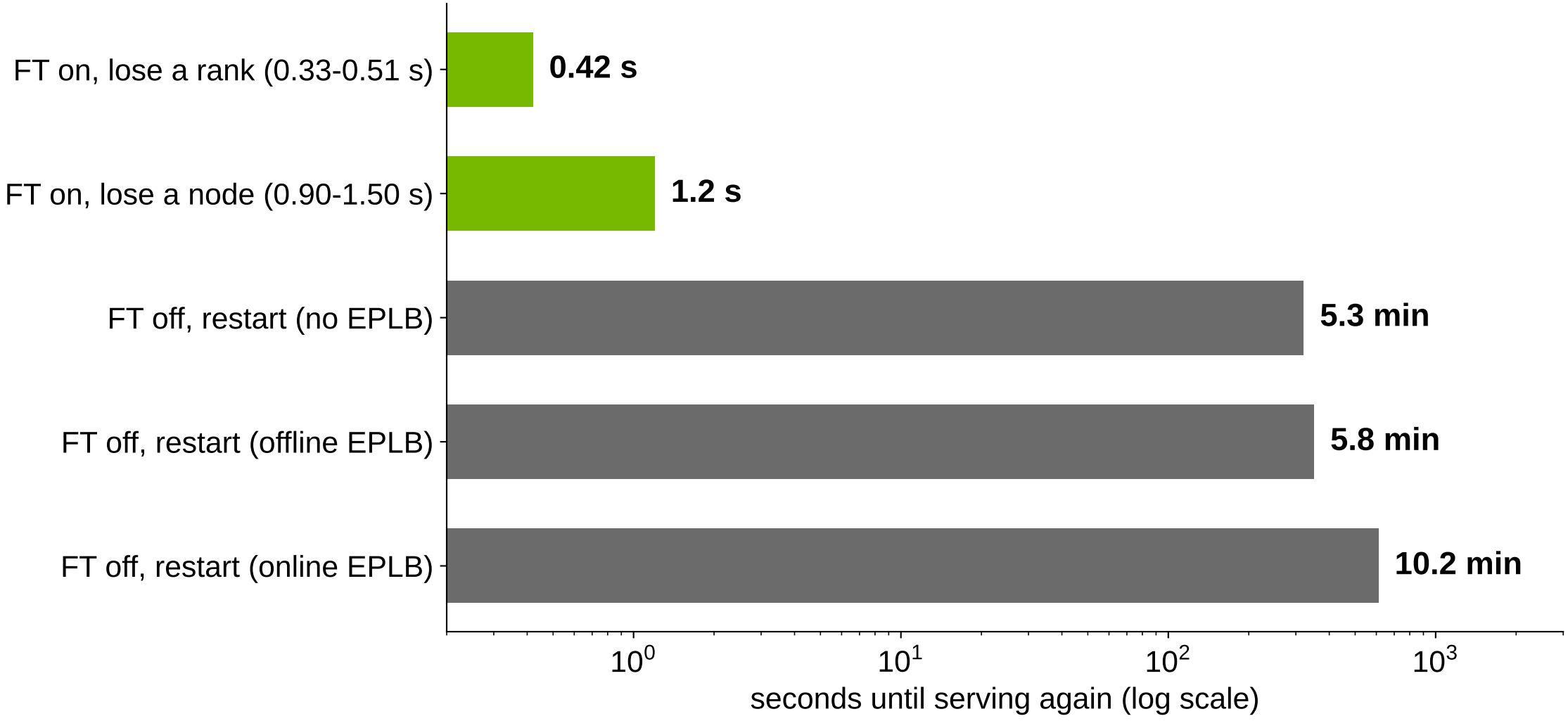
Keep serving when a GPU or node fails:
survivors regroup in about a second on the same CUDA graphs

Mid-point review · DeepSeek-R1 NVFP4 · 32 GB200 GPUs

Losing one GPU stops the whole deployment

LEAD

Without FT, one rank failure means minutes down; with FT, about a second.



FT off: warm restart in the same allocation, RESTART-MODES 8134113 (n=1 per mode). FT on: failure to first step after recovery, v2 final kills 8115497 (bar = median, label = range).

Existing solutions

MOCK

(to fill: vLLM, SGLang, Dynamo restart, ...)

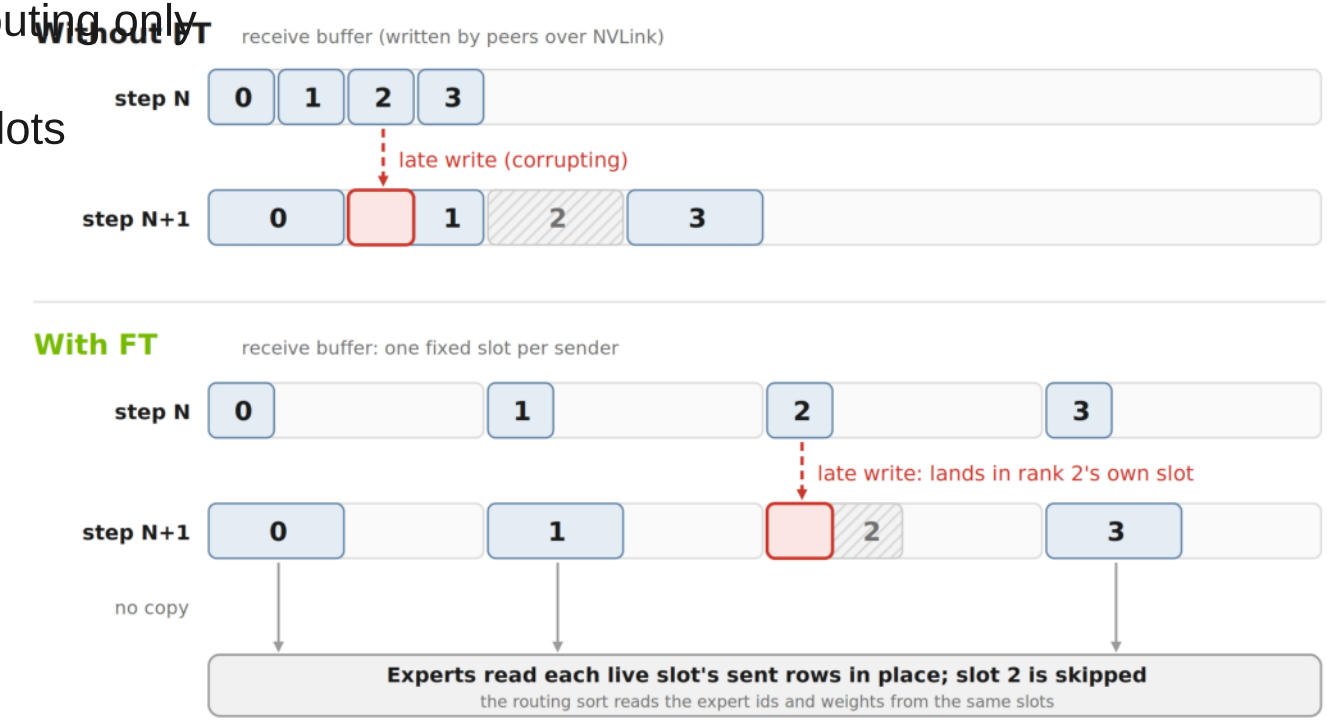
Problem scoping

- In scope: a rank or a whole node fails during serving (crash or hang)
  - DeepSeek-R1 NVFP4, 32 GB200 GPUs, attention DP + 32-way EP, aggregated serving
  - no EPLB, offline EPLB and online EPLB
- Goal: serve again in ~1 s at reduced capacity; FT costs <1% while healthy
- Out of scope (MVP): rejoining a replacement (N-1 → N), other models,
  - disaggregated serving (first feature after the midpoint), losing rank 0

Core principles

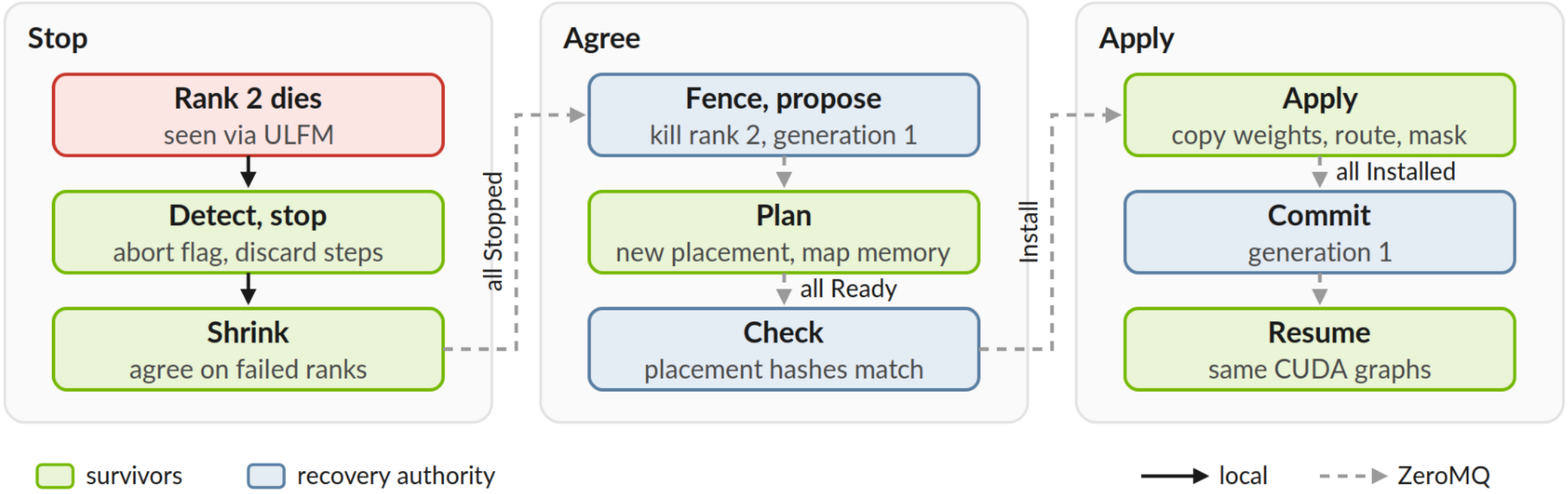
Recovery changes contents, never shapes: the CUDA graphs keep running.

- Fixed sizes and addresses for every table the graph reads
- Recovery rewrites the live-rank mask and expert routing only
- Lost experts restored from host copies into ghost slots
  - ghost slots take no GPU memory until used



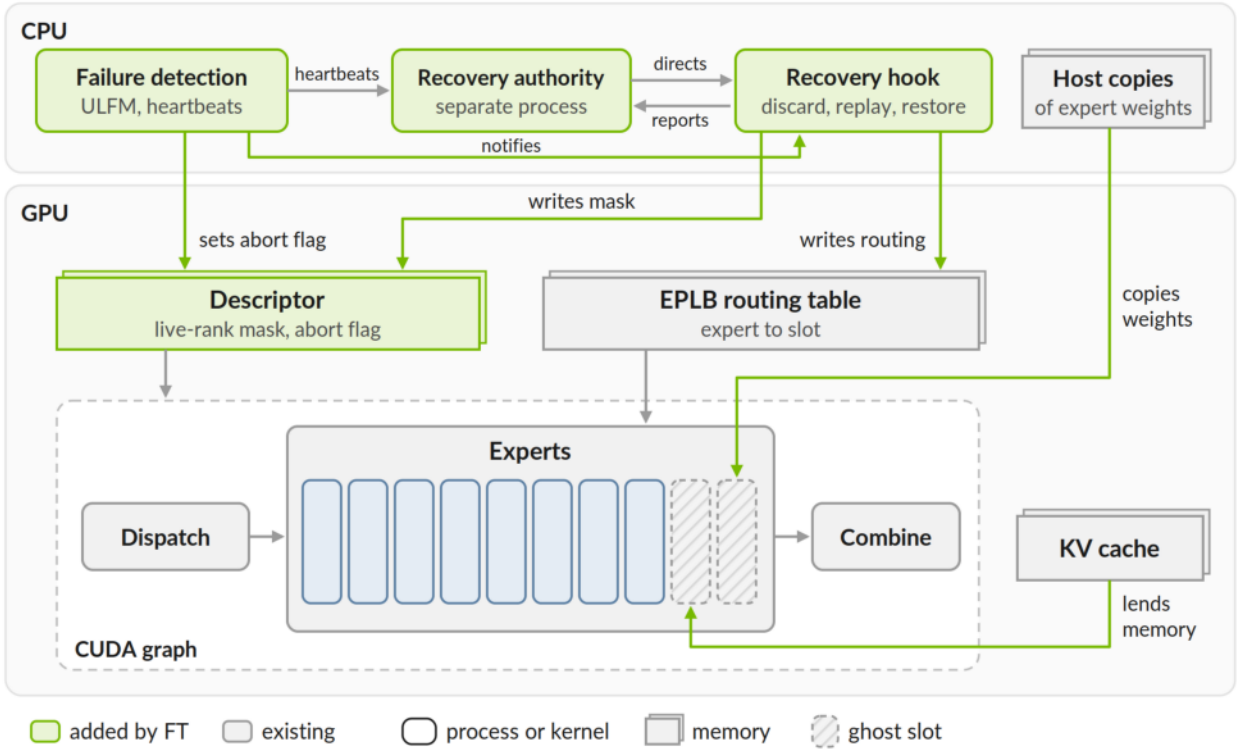
Recovery: stop → agree → apply

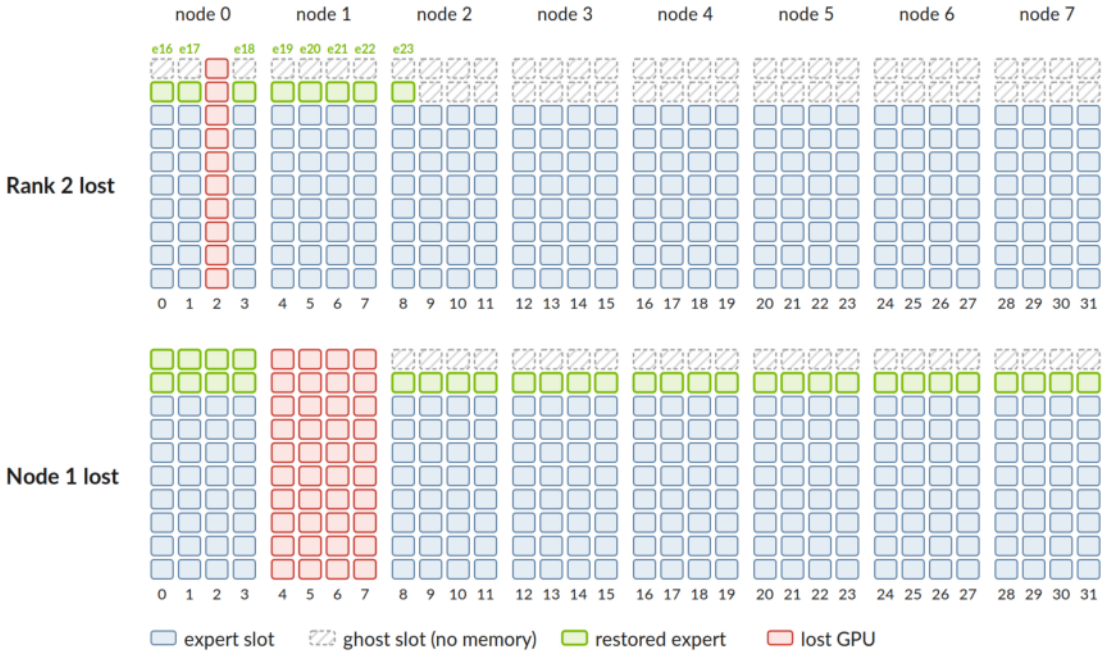
Survivors detect, agree on who is lost, apply the new placement, and resume.



What gets updated

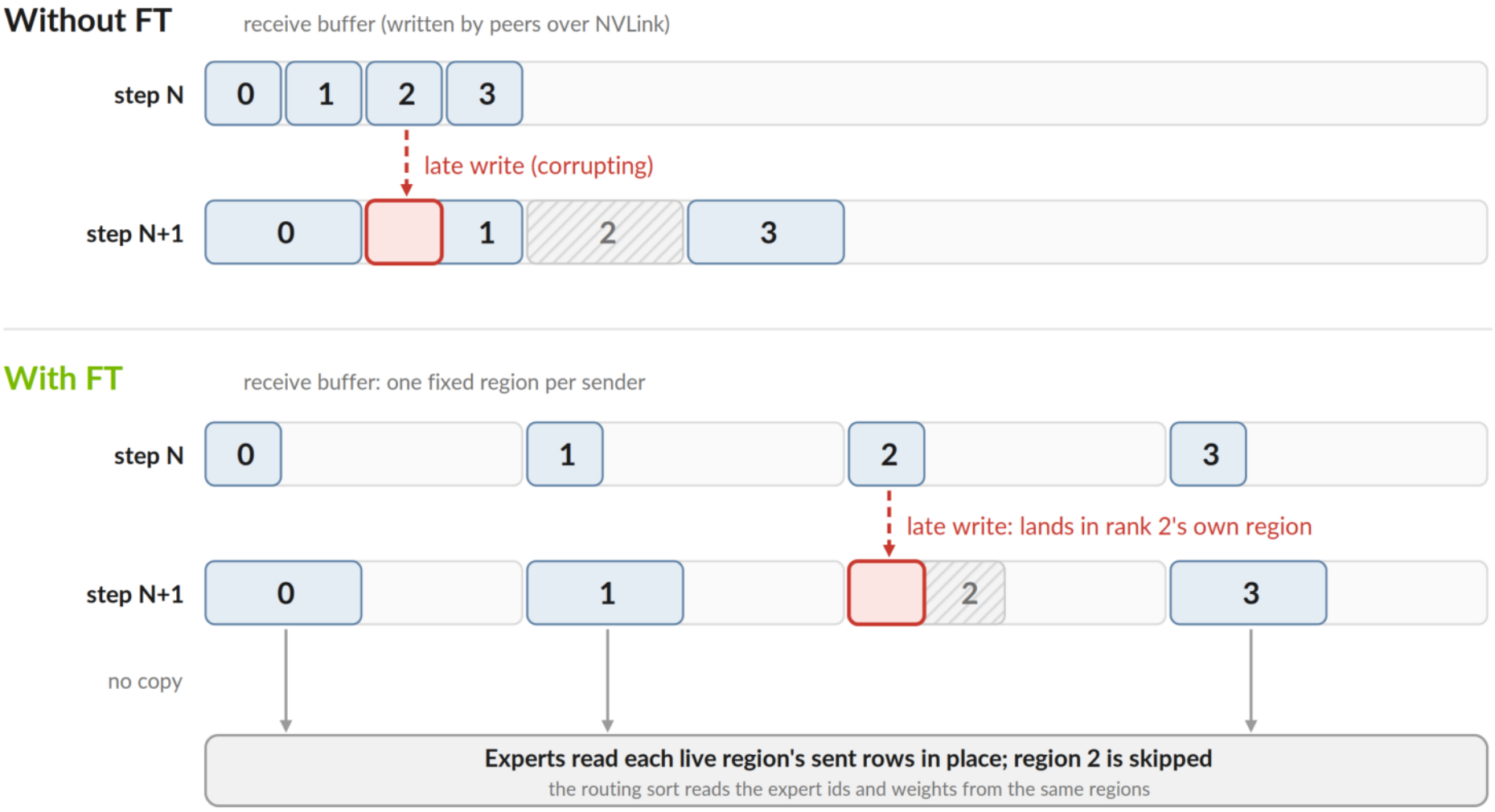
Mask, membership, EPLB placement; lost experts spread over survivors.





Late writes from a dying rank

A failed rank is always fenced before survivors resume.



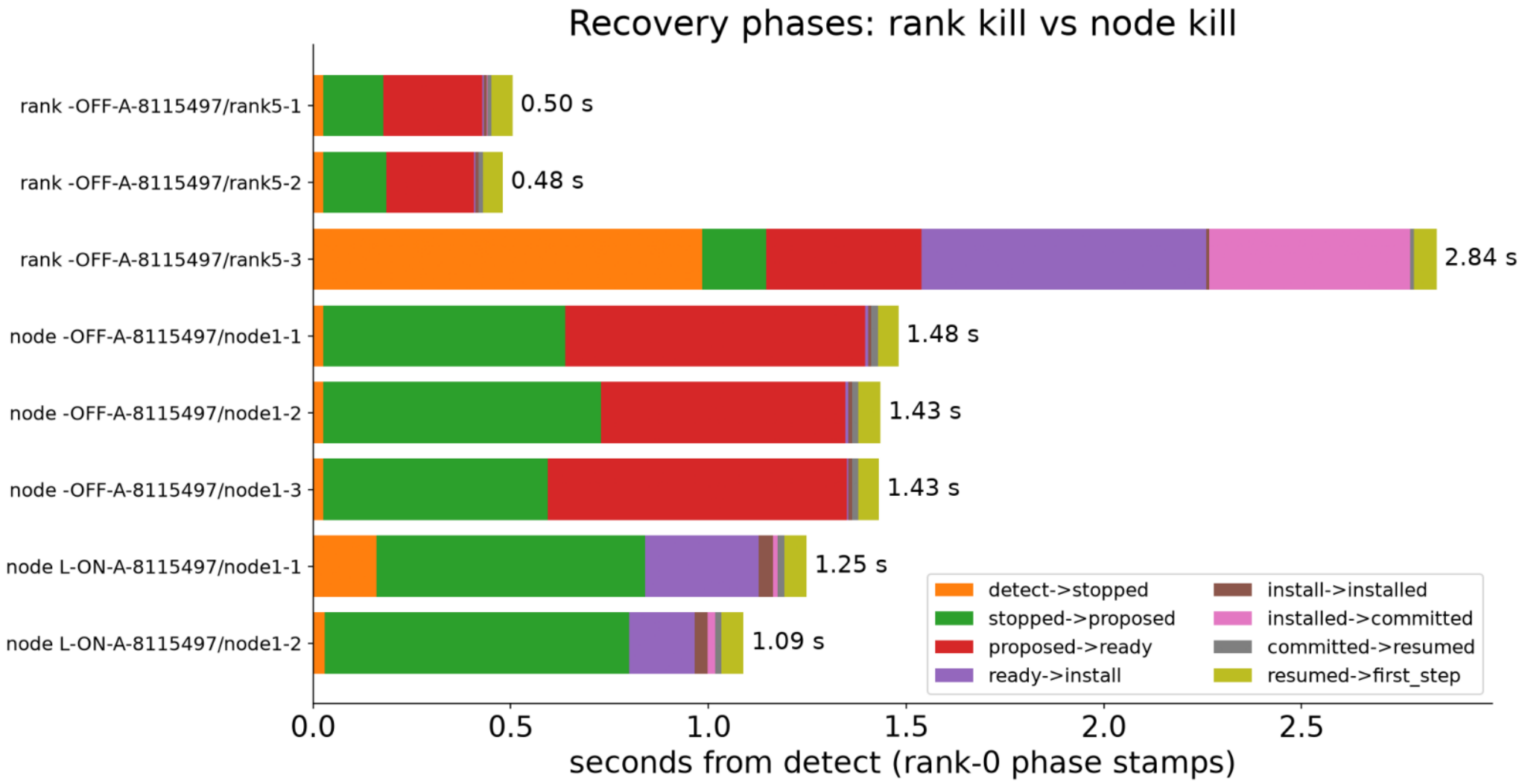
Setup

- DeepSeek-R1 0528 NVFP4, 32 GB200 GPUs (8 nodes × 4), attention DP + EP32, MTP-1, overlap
- TRT-LLM GB200 R1 recipe; benchmark_serving, 1k in / 1k out, 8-512 requests per GPU
- 3 modes: no EPLB, offline EPLB, online EPLB
- FT off = main's code path; FT on = same config + FT
- Each mode on one fixed node set; FT off/on alternating (ABBA); ≥4 repeats; 95% CI
- Failures: process kills (rank, node, frozen rank) at a fixed decode step

Where the recovery second goes

LEAD

Detection and agreement are fast; planning dominates after a node loss.

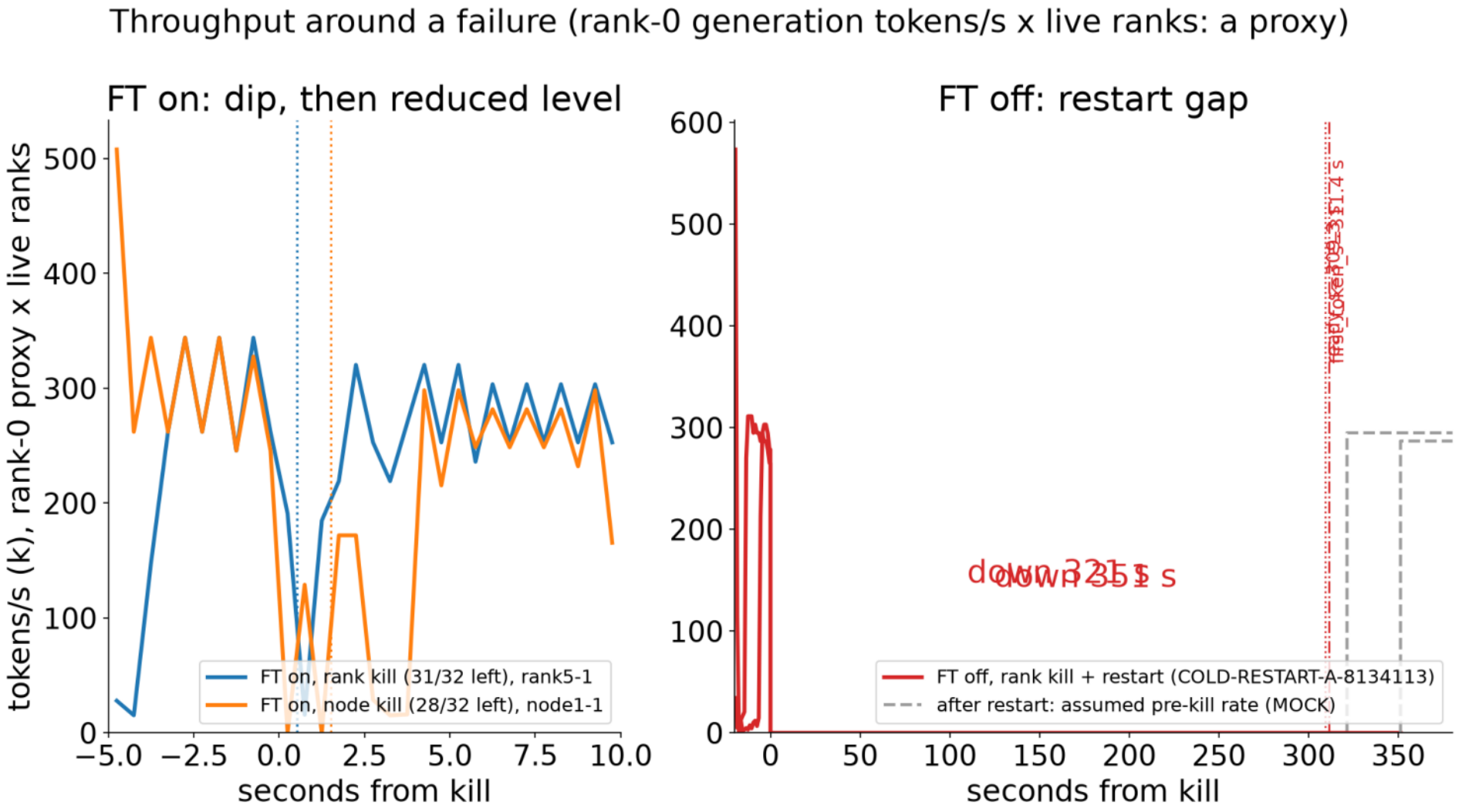


v2 final kills 8115497 + RESTART-MODES 8134113; MOCK parts labelled in the figure. Replaced by the final rerun.

Throughput around a failure

LEAD

A ~1 s dip, then serving continues; without FT, minutes of nothing.

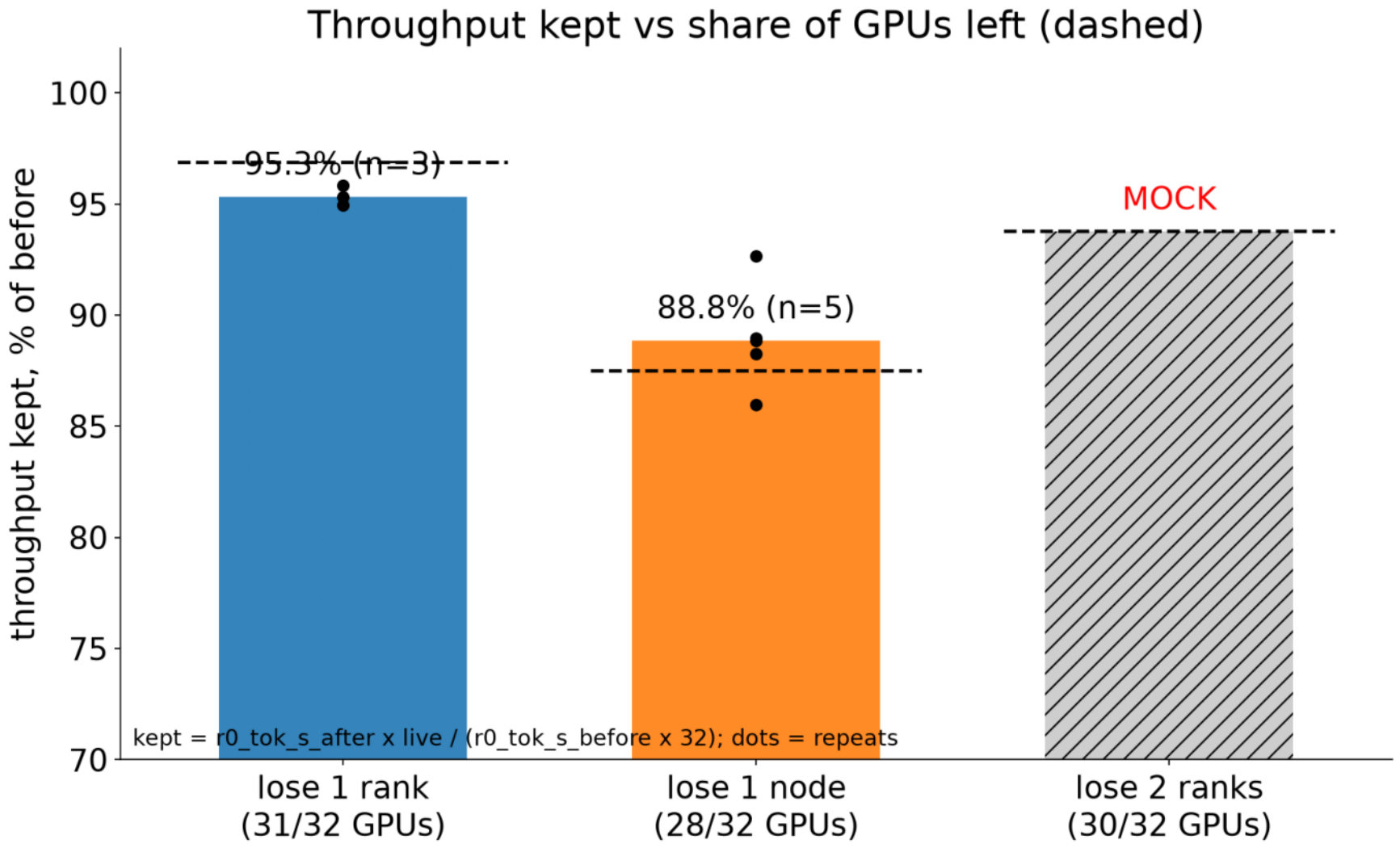


v2 final kills 8115497 + RESTART-MODES 8134113; MOCK parts labelled in the figure. Replaced by the final rerun.

Capacity after losses: N → N-1 → N-k

LEAD

Throughput tracks the share of GPUs left.

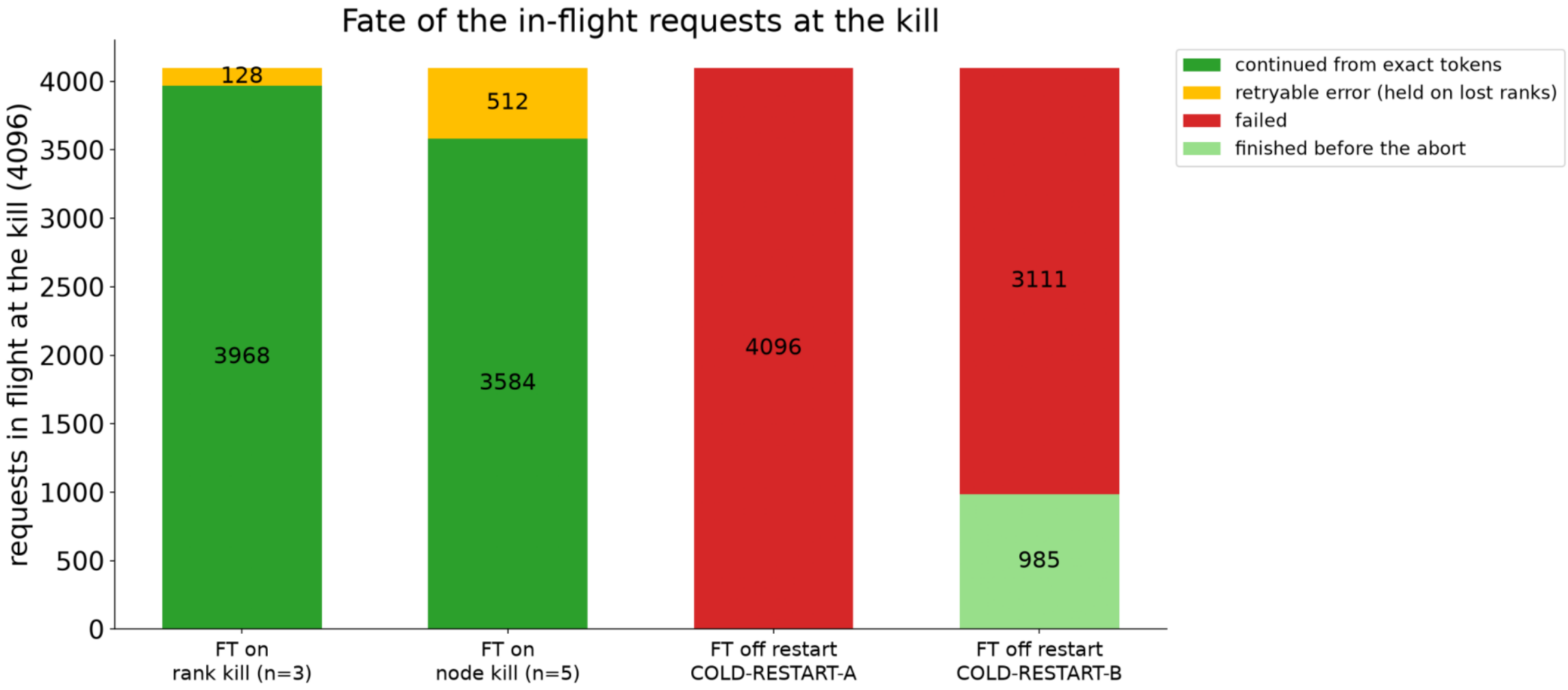


v2 final kills 8115497 + RESTART-MODES 8134113; MOCK parts labelled in the figure. Replaced by the final rerun.

What happens to in-flight requests

LEAD

Most continue; requests on the lost rank get a retryable error; FT off fails all.

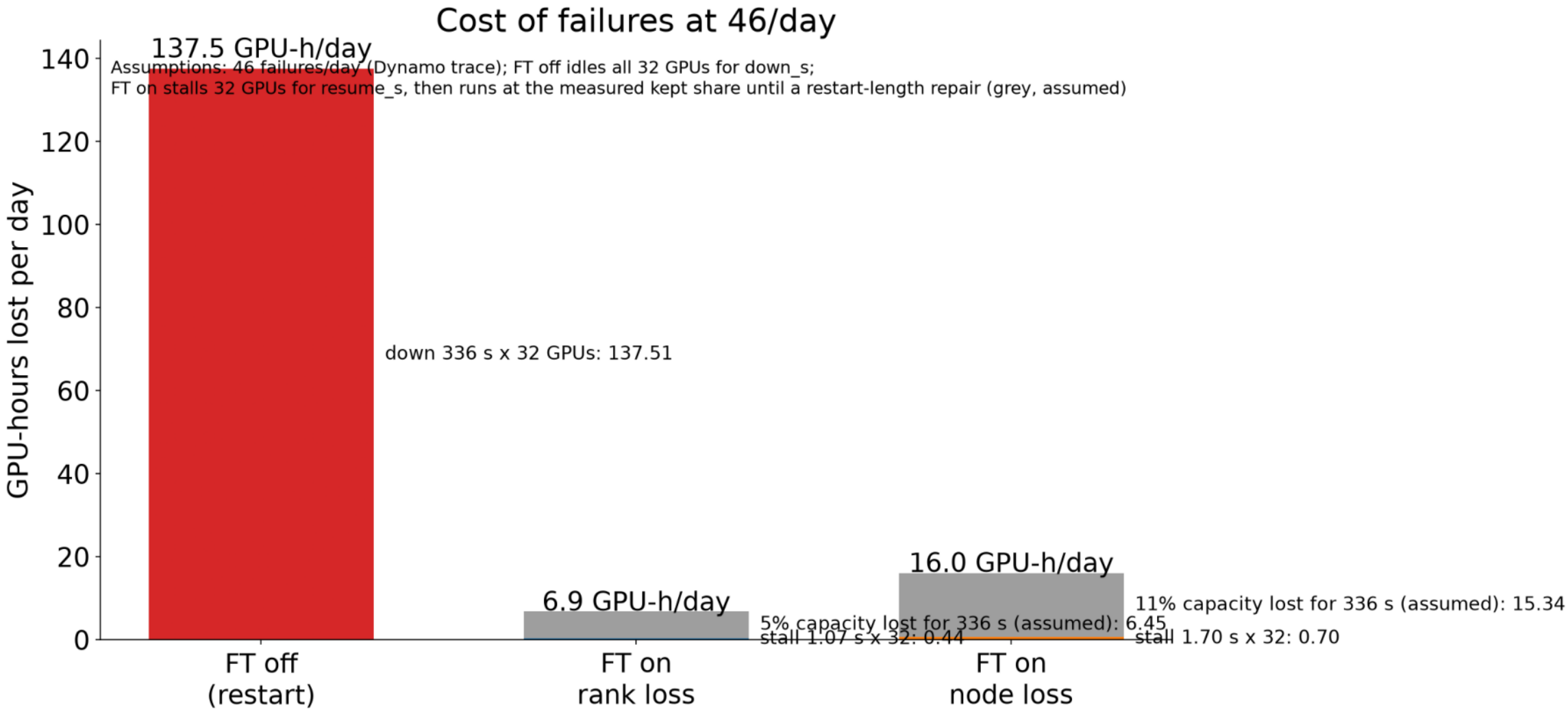


v2 final kills 8115497 + RESTART-MODES 8134113; MOCK parts labelled in the figure. Replaced by the final rerun.

Why it matters: cost per day

LEAD

At 46 failures/day, minutes vs seconds of downtime.



v2 final kills 8115497 + RESTART-MODES 8134113; MOCK parts labelled in the figure. Replaced by the final rerun.

Cost while healthy: FT on vs FT off

INDICATIVE

Whole-run throughput cost per cell; final slide shows the estimate ± 95% CI.

	8/GPU	32/GPU	128/GPU	512/GPU
no EPLB	+2.4%	+0.5%	<1	<1
offline EPLB	+1.7%	+1.8%	<1	–
online EPLB	–	+2.0%	+1.6%	+1.4%

Mechanisms found:

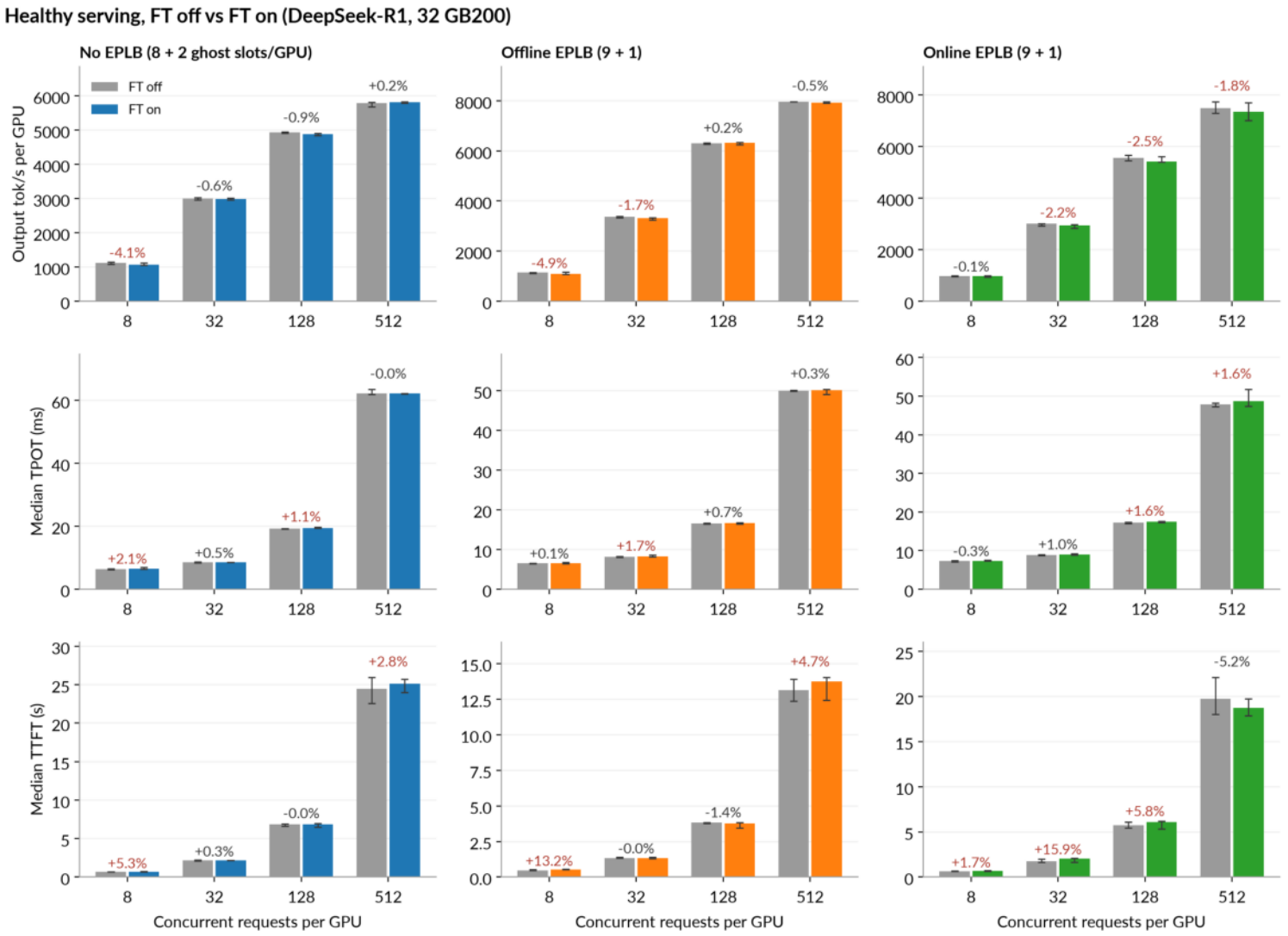
- prefill steps slower
  - – online +15% at 32
- more stalls at 8/GPU
  - – 3.6× rate with FT
- a2a kernels at 32
  - – no EPLB, decode

v2 final runs (Oct 21), whole-window tokens/s, mean over 3-4 node groups (CLAIMS 149). '<1' = within 1% on every metric. Replaced by the final rerun.

Across loads and modes

INDICATIVE

Throughput, TPOT and TTFT, FT off vs FT on.



v2 final runs; replaced by the final rerun.

When does FT pay off?

LEAD

In GPU-hours, FT breaks even near 1 failure/day at ~0.4-0.7% healthy cost; the bigger win is failed requests and minutes of SLO violation.

Failures per day	0.5	1	2	5
GPU-h lost to restarts (no EPLB)	1.4	2.9	5.7	14.3
GPU-h lost to restarts (online EPLB)	2.7	5.4	10.9	27.2
GPU-h FT costs per day at 0.5% healthy cost	3.8	3.8	3.8	3.8
GPU-h FT costs per day at 1.0% healthy cost	7.7	7.7	7.7	7.7
Requests failed per failure	FT off: 3,111-4,09(	FT on: ~128	(rank loss)	

Restart = 32 GPUs x warm restart (321-611 s, RESTART-MODES 8134113, n=1); healthy cost = 32 x 24 h x cost.
Failure rates are estimates (Dynamo trace of ~150 mostly <=8-GPU replicas, extrapolated).

Status: done, risks, asks

Done

- Design + code (rank/node loss, ghost slots, fencing)
- Unit, GPU, integration tests; kills pass
- Recovery ~0.5-1.4 s, 0 client failures

Risks

- Healthy cost >1% in some cells (being fixed)
- ULFM Open MPI needed in the container
- Large PR stack (~14k lines, half tests)

Asks

- PR split and review order
- Config via LlmArgs before merge?
- Who owns ULFM in the image?

Summary

- Problem: one failure takes a 32-64 GPU deployment down for minutes
- Design: survivors regroup in ~1 s on the same CUDA graphs; lost experts restored from host
- Recovery: 0.3-0.5 s (rank) / 0.9-1.5 s (node) vs 5-10 min restart; 89-95% throughput kept
- Healthy cost: being driven below 1% in every cell (final numbers pending)

Next steps

- Upstream: fixes to main first, then the FT core PRs
- Disaggregated serving
- Rejoining a replacement while serving (N-1 → N)
- Other models (GLM-5.x), real host loss, CI with ULFM Open MPI

Questions & discussion